

Quiz 1

*Total Marks: 25**24th August 2026*

1. State whether the following statements are TRUE or FALSE and justify your answers. Answers without justification will fetch zero marks. [5 × 1 marks]
- (a) If a language L is regular, so is the language $L' = \{w : w^{|w|} \in L\}$
 - (b) If two states have the same ε -closure, merging them in ε -NFA will change the language accepted by the NFA.
 - (c) The language over $\Sigma = \{0, 1 \dots 9\}$ that accepts strings in which the final digit has not appeared before is regular.
 - (d) If a language L can be represented by a DFA with n states, L has at most n equivalence classes.
 - (e) If two regular languages $L(P), L(Q)$ are disjoint, then $L(P^*), L(Q^*)$ are also disjoint.

Solution

(a) **True**

Let $f : Q \rightarrow Q$ be a function such that $f_w(q)$ represents the state that the DFA ends up in after reading w starting from q . We need to check if $f_w^{|w|}(q_0) \in Q_f$. In the new DFA, we can encode all possible transformations $f_w(q) \forall q \in Q, w \in \Sigma^*$ since there are only n^n total transformations from $Q \rightarrow Q$. For the exponent, notice that $q_0, f_w(q_0), f_w^2(q_0) \dots$ will eventually become periodic. So to maintain the number of times w is read, we just need to check it modulo a number (what?) making the number of states finite.

(b) **False**

If two states have the same ε -closure, for any alphabet, they will have the same set of next states and hence you cannot find a word that distinguishes both states.

(c) **True**

Construct a NFA where each state represents the digits seen so far and each one has transitions to final state (which has no transitions) with all digits not in that state.

(d) **True**

If L had more than n equivalence classes, by PHP, there exists a state in the DFA that represents at least 2 equivalence classes which is not possible.

(e) **False**

Both accept empty word.

(or)

Let P be a language that contains only the word ab and Q similarly contains only the word $abab$.

Rubrics

- **1 mark** for correct answer with proper justification.
- **0.5 marks** for correct answer with somewhat reasonable justification
- **0 marks** if wrong answer/justification

2. Let L be a regular language, and let $\text{sc}(L)$ denote the number of states of its minimal DFA. *Jaffe's pumping lemma* states that L is regular if and only if there exists n such that every $y \in \Sigma^*$ with $|y| = n$ can be written $y = uvw$ (with $v \neq \varepsilon$) so that for every continuation $z \in \Sigma^*$ and every $t \geq 0$,

$$uvwz \in L \iff uv^t wz \in L.$$

- (a) **Prove:** for any regular language L , we can choose n as $\text{sc}(L)$ and satisfy Jaffe's pumping property. [4 marks]
- (b) Give an example of a language L' such that $\text{jp}(L') < \text{sc}(L')$. [1 mark]
- (c) Let $A = \langle Q, \Sigma, \delta, q_0, F \rangle$ be the minimal DFA for language L . Characterise all A such that

$$\text{jp}(L_A) < \text{sc}(L_A)$$

Your characterisation should be in a **iff** formulation involving Q and δ and you need to prove both implications. [6 marks]

Solution

- (a) We begin by proving $n = \text{sc}(L) = |Q|$ always works

Take any $y \in \Sigma^*$ with $|y| = n$. Consider $n + 1$ states visited while reading y : $q_0, \delta(q_0, y_1), \delta(q_0, y_1 y_2), \dots, \delta(q_0, y)$

By PHP, two of these $n + 1$ states must be equal. Thus $\exists 0 \leq i < j \leq n$ such that

$$\delta(q_0, a_1 \dots a_i) = \delta(q_0, a_1 \dots a_j)$$

We set

$$u = a_1 \dots a_i$$

$$v = a_{i+1} \dots a_j$$

$$w = a_{j+1} \dots a_n$$

$\therefore i < j$, $v \neq \epsilon$ and $y = uvw$. Additionally $\delta(q_0, u) = \delta(q_0, uv)$

So, v is at a loop at the state reached after reading u . So for every $t \geq 0$, $\delta(q_0, uvw) = \delta(q_0, uv^t w)$. So for all suffixes $z \in \Sigma^*$, $\delta(q_0, uv^t wz) = \delta(q_0, uvwz)$ which mean $uvwz \in L \equiv uv^t wz \in L$. Since y was arbitrary, $n = \text{sc}(L)$ satisfies Jaffe's property.

- (b) $L' = \{w \in \{0, 1\}^* : |w| \text{ the first symbol is } 0\}$ over $\Sigma = \{0, 1\}$

The minimal DFA has 3 states q_0, q_1, q_2 where q_0 is the initial state and q_1 is the

		0	1
accepting state. The transition function	q_0	q_1	q_2
	q_1	q_1	q_1
	q_2	q_2	q_2

so $\text{sc}(L') = 3$. For every $y \in \{0, 1\}^2$, reading y from q_0 returns to q_0 . So we can choose $u = \epsilon, v = y, w = \epsilon$ and v is a non-empty loop at q_0 . So $n = 2$ satisfies Jaffe's property $\Rightarrow \text{jp}(L') \leq 2$.

$n = 1$ does not work. For $y = 0$ the only the only possible decomposition is $u = \epsilon, v = 0, w = \epsilon$. Taking $z = \epsilon$ and $t = 0$ would need $0 \in L' \equiv \epsilon \in L'$. So $\text{jp}(L') = 2$

- (c) Let $A = (Q, \Sigma, \delta, q_0, F)$ be minimal, $n = |Q| = \text{sc}(L_A)$. For $y = a_1 \cdots a_m$ write $q_0 = q_0$, state sequence for its run is $q_0, q_1 \dots q_m$ where $q_k = \hat{\delta}(q_0, a_1 \cdots a_k)$

Claim: $\text{jp}(L_A) < \text{sc}(L_A) \iff$ every Hamiltonian word of A has a “good pair” where:

- a word $y = a_1 \cdots a_{n-1} \in \Sigma^{n-1}$ is **Hamiltonian** if its run on A visits all n states of Q , each exactly once
- for such y , a pair $0 \leq i < j \leq n - 1$ is a “**good pair**” if, writing $v = a_{i+1} \cdots a_j$ and $w = a_{j+1} \cdots a_{n-1}$,

$$\delta^*(q_i, v^t w) = q_{n-1} \quad \text{for every } t \geq 0 \quad (*)$$

Step 1: Reduce to checking one word length

(A) Monotonicity of Jaffe's Property: If Jaffe's property $J(n)$ holds at length n , it must hold at $n + 1$ as well

Reasoning: Take $y \in \Sigma^{n+1}$, split $y = y'c$ with $|y'| = N$. Apply $J(n)$ to y' : get $y' = uvw'$ with $\forall z', t: uvw'z' \in L \Leftrightarrow uv^t w'z' \in L$. This holds for all z' , so plug in $z' = cz$ for arbitrary z :

$$\forall z, t: \quad uv(w'c)z \in L \iff uv^t(w'c)z \in L.$$

So $y = uvw$ with $w = w'c$ satisfies the property.

$J(n)$ hold (part a). With that and monotonicity we get $\text{jp}(L_A) < n \iff J(n-1)$. So we only need to characterise when $J(n-1)$ holds, all words of length $n-1$ admits a valid uvw split.

Step 2: Translate the semantic condition into a structural one

(B) Minimality $\Rightarrow$ states separated by continuations

Since A is minimal, for $p, q \in Q$:

$$p = q \iff \forall z \in \Sigma^* : \delta^*(p, z) \in F \iff \delta^*(q, z) \in F.$$

(C) Fix y of length $n - 1$, and a split point $i < j$ giving u, v, w as above. Then

$$\forall z, t : uvwz \in L \iff uv^t wz \in L$$

holds **iff** $(*)$ holds.

Reasoning:

$$uv^t wz \in L \iff \delta^*(q_0, uv^t wz) \in F \iff \delta^*(\delta^*(q_i, v^t w), z) \in F,$$

using $\delta^*(q_0, u) = q_i$. Similarly $uvwz \in L \iff \delta^*(q_{n-1}, z) \in F$ (the $t = 1$ case just retraces the original path, landing on q_{n-1}). So the equivalence for all z says exactly, by B, that $\delta^*(q_i, v^t w) = q_{n-1}$ for every t .

So, y has a valid uvw split $\equiv y$ has a good pair (i, j) .

(D) If $y \in \Sigma^{n-1}$ is not Hamiltonian, it automatically has a good pair.

Reasoning: Its state sequence $q_0, \dots, q_{n-1}$ has n entries but doesn't hit all n states. So by PHP, some state repeats: $q_i = q_j$ for some $i < j$. Then $v = a_{i+1} \dots a_j$ satisfies $\delta^*(q_i, v) = q_j = q_i$, so v loops at q_i : $\delta^*(q_i, v^t) = q_i$ for all $t \geq 0$. This means

$$\delta^*(q_i, v^t w) = \delta^*(q_i, w) = q_{n-1} \quad \text{for every } t,$$

so $(*)$ holds and (i, j) is a good pair.

So the only words that can possibly break $J(n - 1)$ are the Hamiltonian ones (state seq not repeating so the repeated state thing we did in a doesn't hold anymore)

Step 4: Proof of the iff

By Step 1, $jp(L_A) < sc(L_A) \iff J(n - 1)$ holds $\iff$ every $y \in \Sigma^{n-1}$ has a good pair.

($\Leftarrow$) Suppose every Hamiltonian word has a good pair. Take any $y \in \Sigma^{n-1}$. If y is non-Hamiltonian, it has a good pair by D. If y is Hamiltonian, it has one by assumption. Either way y has a good pair, so by C, y has a valid uvw -split. As y was arbitrary, $J(n - 1)$ holds, so $jp(L_A) < n = sc(L_A)$.

($\Rightarrow$) Suppose $jp(L_A) < sc(L_A)$, so $J(n - 1)$ holds. Let y be any Hamiltonian word. Since $J(n - 1)$ holds, y has some valid split $y = uvw$ with $v \neq \varepsilon$, corresponding to indices $i < j$. According to C, this is exactly a good pair (i, j) for y . Since y was an arbitrary Hamiltonian word, every Hamiltonian word has a good pair.

Rubrics

- (a) 2 marks for reasoning about the run, using PHP
2 marks for the correct decomposition of the word and reasoning about why it follows Jaffe's pumping property. 0.5 marks have been deducted for mistakes including (but not limited to) not drawing start states for automata, saying the $uv^i w$ ends in an accepting state.
- (b) 1 mark for mentioning the example and **jp, sc**. 0.5 marks for only mentioning the example. No marks have been given where two answers have been given and at least one of them is incorrect (please do not raise any discrepancies about this).
- (c) 6 marks for the correct **iff** formulation (1 for characterisation, 2 for backward direction, 3 for forward direction of the proof)
3 marks have been given for sufficient but not necessary characterisation (0.5 for characterisation, 2.5 for the proof (1 for backward, 1.5 for forward direction))
Again, 0.5 marks have been deducted for mistakes like gap in the reasoning.

3. Recall that we defined regular expressions in class. Let us look at a special class of regular expressions called *star-free expressions*. Inductively these are defined as follows.

A star-free expression over an alphabet Σ is built from primitives $\epsilon, \emptyset$ and $a \in \Sigma$ using union (+), concatenation ($\cdot$) and complementation (denoted $\overline{R}$ for a star free expression R). Note that Kleene star is not allowed, as the name suggests (star-free).

A language L is called star-free if there is a star-free expression R such that $L = L(R)$. Prove or disprove whether the following languages are star-free.

- (a) Σ^* [**1 mark**]
- (b) $(ab)^*$ [**4 marks**]

Solution

Both the languages are star-free.

- (a) $\overline{\emptyset}$ or $R + \overline{R}$ where R is any star-free expression
- (b) $(ab)^*$ can be captured with the following constraints: A word should 1. start with a , 2. end with b , 3. have no 2 consecutive a 's, 4. have no consecutive 2 b 's.

$$R = \overline{b\overline{\emptyset} + \overline{\emptyset}a + \overline{\emptyset}aa\overline{\emptyset} + \overline{\emptyset}bb\overline{\emptyset}}$$

Rubrics

- (a) 1 mark for correct expression.
- (b) 4 marks for correct expression.

No marks are awarded if the language represented by the language is wrong.

4. Given $\Sigma = \{0, 1\}$ consider the family of languages $L_n = \{ww \mid w \in \Sigma^n\}$. Show that an NFA accepting L_n has $\geq 2^n$ states. [4 marks]

Solution

(d) Let $A = (Q, \Sigma, \delta, q_0, F)$ be an NFA accepting

$$L_n = \{ww \mid w \in \{0, 1\}^n\}.$$

For every $x \in \{0, 1\}^n$, define

$$AS(x) = \{q \in Q \mid q \in \hat{\delta}(q_0, x), \hat{\delta}(q, x) \in F\}$$

Since $xx \in L_n$, there exists an accepting run of xx . Hence

$$AS(x) \neq \emptyset$$

for every $x \in \{0, 1\}^n$.

We claim that for any two distinct $x, y \in \{0, 1\}^n$,

$$AS(x) \cap AS(y) = \emptyset.$$

Suppose, for contradiction, that there exists a state

$$q \in AS(x) \cap AS(y).$$

Since $q \in AS(x)$, there is an accepting run for xx which reaches q after reading the first x , and then continues from q while reading x to an accepting state.

Similarly, since $q \in AS(y)$, there is an accepting run for yy which reaches q after reading the first y , and then continues from q while reading y to an accepting state.

We can therefore combine the part of the accepting run for xx that reads x and reaches q with the part of the accepting run for yy that starts at q and reads y . This gives an accepting run for

$$xy.$$

Thus,

$$xy \in L_n.$$

However, since $x \neq y$ and $|x| = |y| = n$, we have

$$xy \notin L_n,$$

because a string in L_n must consist of two identical n -bit words. This is a contradiction.

Hence, the sets $AS(x)$ are pairwise disjoint and non-empty.

There are 2^n different words in $\{0, 1\}^n$. Therefore,

$$|Q| \geq \sum_{x \in \{0, 1\}^n} |AS(x)| \geq 2^n.$$

Thus, every NFA accepting L_n has at least

$$2^n$$

states.